

Product Sentiment Analyzer and Review Dashboard

Team - 2

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Make every review meaningful.

SentimentAI turns customer language into clear signals your team can act on, from the first review to the latest trend.

Analyze a product →

View dashboard

Total products

2

catalog

Total reviews

40

indexed

Positive sentiment

30

reviews

Negative sentiment

10

reviews

Average rating

3.5 / 5

across catalog

72

sentiment
index

↑ 18.4%

positive momentum

Project Description

The **Product Sentiment Analyzer and Review Dashboard** is a web-based application that collects product reviews from e-commerce websites such as Amazon and Flipkart using dynamic web scraping.

Review Collection

The application analyzes the reviews using sentiment analysis techniques and displays the insights on an interactive dashboard. Reviews are classified as **positive, negative, or neutral** using NLP libraries.

Search & Visualization

Users can search for a product, view its reviews, and explore visual sentiment trends through interactive charts, making product research intuitive and data-driven.

Storage & Hosting

The scraped and processed data is stored in a database and presented via a user-friendly interface. The application is hosted using free cloud services for broader accessibility.

Key Features

	Dynamic Review Scraping Scrapes reviews in real-time from major e-commerce sites using Selenium, ensuring up-to-date and accurate data collection.
	Sentiment Classification Analyzes the sentiment of each review using TextBlob (NLP) or VADER, classifying them as positive, negative, or neutral.
	UI for Search & Visualization Enables users to search for products and view sentiment charts and reviews through an intuitive, interactive interface.
	API Integration Backend APIs serve data to the frontend and handle user queries securely, ensuring reliable communication between layers.
	Database Connectivity Stores product data, reviews, and sentiment analysis results in MongoDB or PostgreSQL for persistent and structured data management.
	Visual Analytics Displays sentiment distribution, trends over time, and word frequency using interactive charts for comprehensive data insight.
	Cloud Hosting Deployed using platforms like AWS EC2, Render, and Vercel/Netlify for scalable, globally accessible deployment.

Technologies Used — Interface & Backend

Interface Technologies



React.js

Building the user interface with component-based architecture for a dynamic, responsive experience.

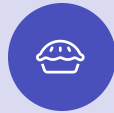


Chart.js / Recharts

Interactive data visualization for rendering sentiment charts, trend graphs, and word frequency displays.



Axios

Handling API communication between frontend and backend with promise-based HTTP requests.

Backend Technologies



Flask / Django

Web framework powering the backend logic, routing, and API endpoint management for the application.



Selenium & BeautifulSoup

Web scraping tools that dynamically extract product reviews from Amazon and Flipkart in real-time.



TextBlob / VADER

NLP libraries performing sentiment analysis and classification of each scraped review.



pandas & NumPy

Data handling and processing libraries used for cleaning, transforming, and aggregating review datasets.

Technologies Used — Database & Deployment

Database

MongoDB Atlas or **PostgreSQL**, accessed via **ElephantSQL** or **Supabase**, is used to store product data, reviews, and sentiment analysis results in a structured and scalable manner.

MongoDB Atlas

NoSQL cloud database for flexible, document-based storage of review data and sentiment results.

PostgreSQL

Relational database accessed via ElephantSQL or Supabase for structured, query-optimized data storage.

Deployment Platforms

Render / AWS EC2

Backend hosting platforms providing scalable compute resources for the Flask/Django application server.

Netlify / Vercel

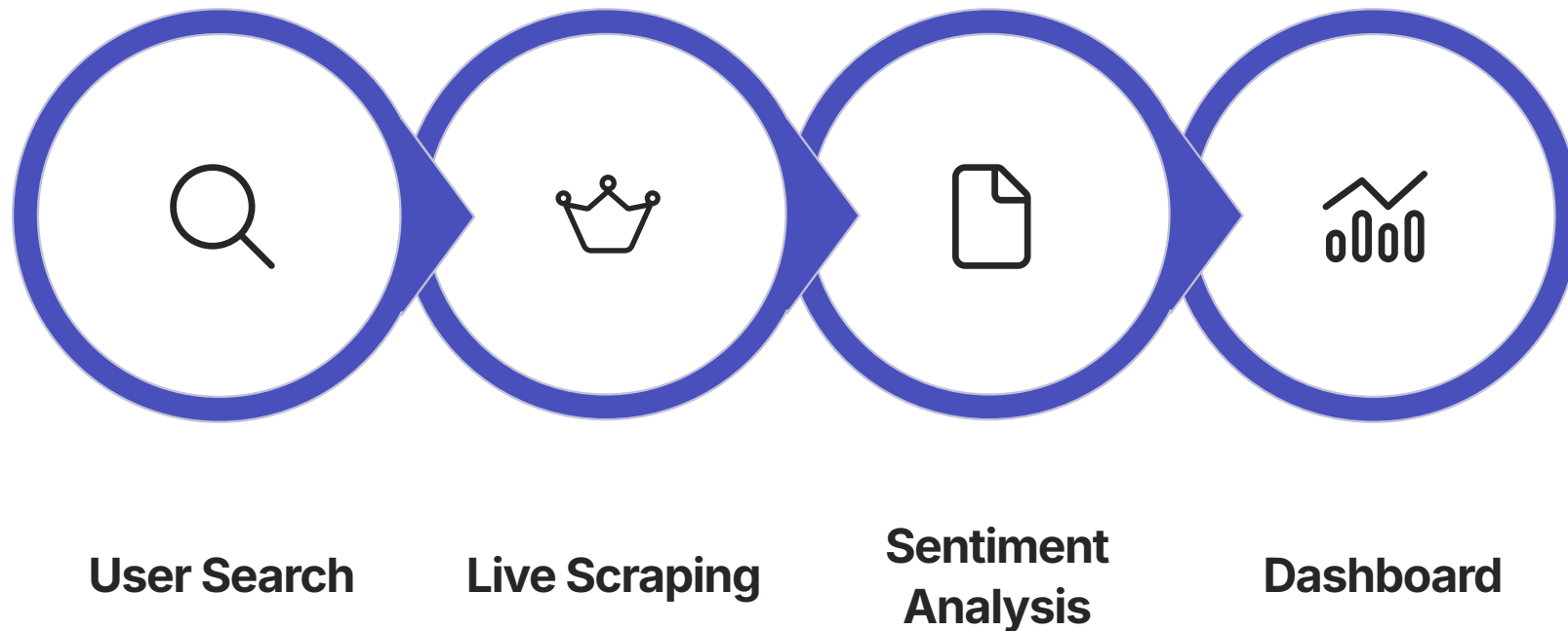
Frontend hosting platforms optimized for React.js deployments with global CDN distribution.

MongoDB Atlas

Cloud database hosting ensuring persistent, globally accessible data storage for the application.

This combination of tools enables a **fully cloud-hosted, scalable, and accessible** application architecture.

System Workflow & Visual Analytics



Data Flow

The user searches for a product through the interface, triggering the backend to scrape live reviews from **Amazon and Flipkart** using Selenium and BeautifulSoup. Each review is processed through the sentiment analysis engine and classified as **positive, negative, or neutral**.

Dashboard Insights

Results are stored in the database and served to the frontend through **secure APIs**. The dashboard visualizes **sentiment distribution, sentiment trends over time, and frequently used terms** through interactive charts, giving users a clear picture of overall product perception.

Outcomes

1 Product Review Insights

Helps users understand the general sentiment around a product **before purchase**, enabling more informed buying decisions based on aggregated customer opinions.

2 Business Intelligence

Enables companies to **identify strengths and weaknesses** based on customer feedback, supporting strategic product development and marketing decisions.

3 Data-Driven Reports

Provides visual data insights such as **rating trends and frequently used terms**, transforming raw review text into actionable business intelligence.

4 Review Monitoring

Offers **real-time monitoring and historical tracking** of product sentiment, allowing businesses to respond quickly to shifts in customer perception.

- ❏ This project demonstrates how **web scraping, NLP-based sentiment analysis, and interactive dashboards** can be combined into a practical, cloud-hosted business intelligence tool.

Sentiment AI

WORKSPACE

Home

Dashboard

Products

Product Search

Reviews

INTELLIGENCE

Sentiment Analysis

Analytics

Trends

Scraper

Command center

● Live workspace

Total products

2

catalog

Total reviews

40

indexed

Positive sentiment

30

reviews

Negative sentiment

10

reviews

Average rating

3.5 / 5

across catalog

Sentiment distribution



● Positive 30 ● Neutral 0 ● Negative 10

Sentiment balance

